HCMUS
Viet Nam National University
Ho Chi Minh City
University of Science

CSC10006 – Introduction to Database

Database Definition

(Defining the database schema using SQL)

Bộ môn HTTT - Khoa CNTT ĐH KHTN

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1 Objectives

Use SQL to implement a relational data model on a specific DBMS: create databases and tables, declare integrity constraints (primary keys, foreign keys, domain/value constraints, etc.), and populate the database with data.

After completing this exercise, students will be able to:

- Execute a SQL script and inspect the execution results in the Query Analyzer tool.
- Use the syntax to create and drop databases.
- Use table-creation statements, including:
  - Creating a table without a primary key
  - Creating a table with a primary key
  - Creating a table with foreign keys
  - Creating a table whose primary key consists of multiple attributes (composite primary key)
- Modify table schemas, including:
  - Adding/removing constraints: primary key, foreign key, domain/value constraints, UNIQUE, and NULL/NOT NULL constraints
  - Adding/removing columns and changing a column's data type
  - Applying additional options when defining foreign keys
- Use supporting statements/tools, such as viewing a table's schema, listing the attributes that form a table's primary key, and listing a table's foreign keys.
- Insert data into a table successfully using multiple methods.
- Insert data for different data types (strings, Unicode strings, dates, etc.).
- Update data.
- Query and delete data.

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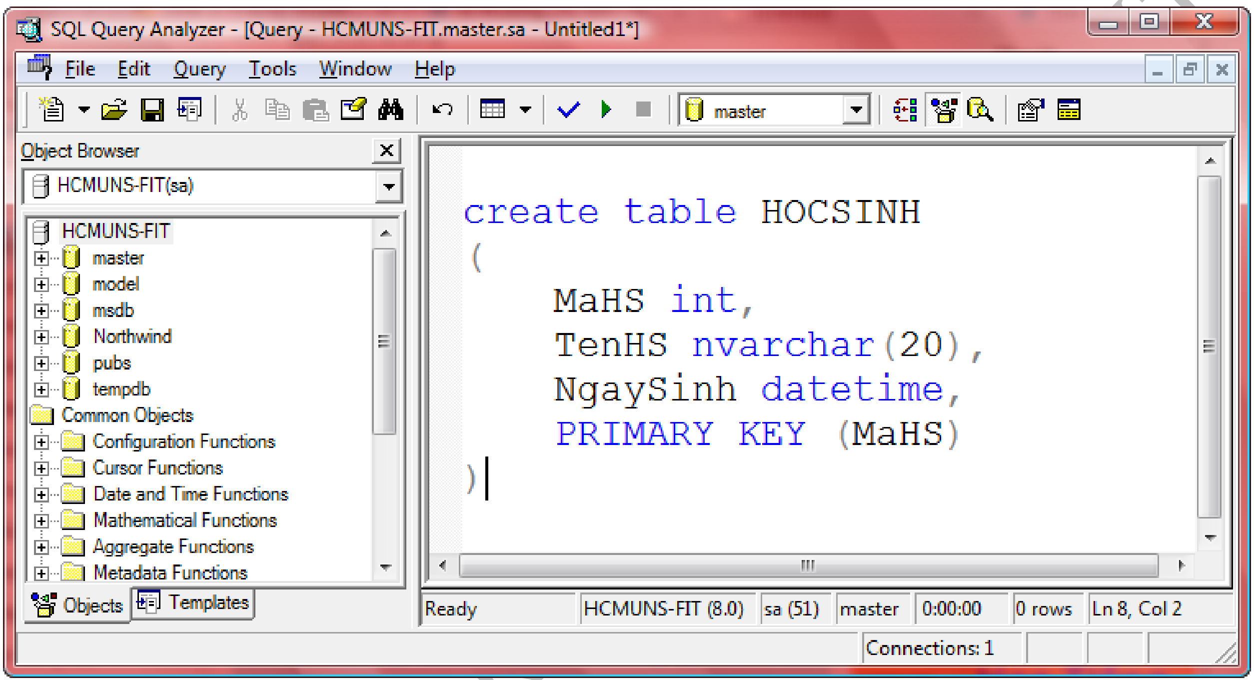
2 Detailed Instructions

2.1 Constructing the Database schema

2.1.1 Execute a Script in Query Analyzer

Assume that we have the following script to create the **HOCSINH** table in the **master** database (you do not need to understand the details of this script yet):



To execute this SCRIPT in Query Analyzer, proceed as follows:

<u>**Step 1**</u>: Select (highlight) the SCRIPT block you want to execute.

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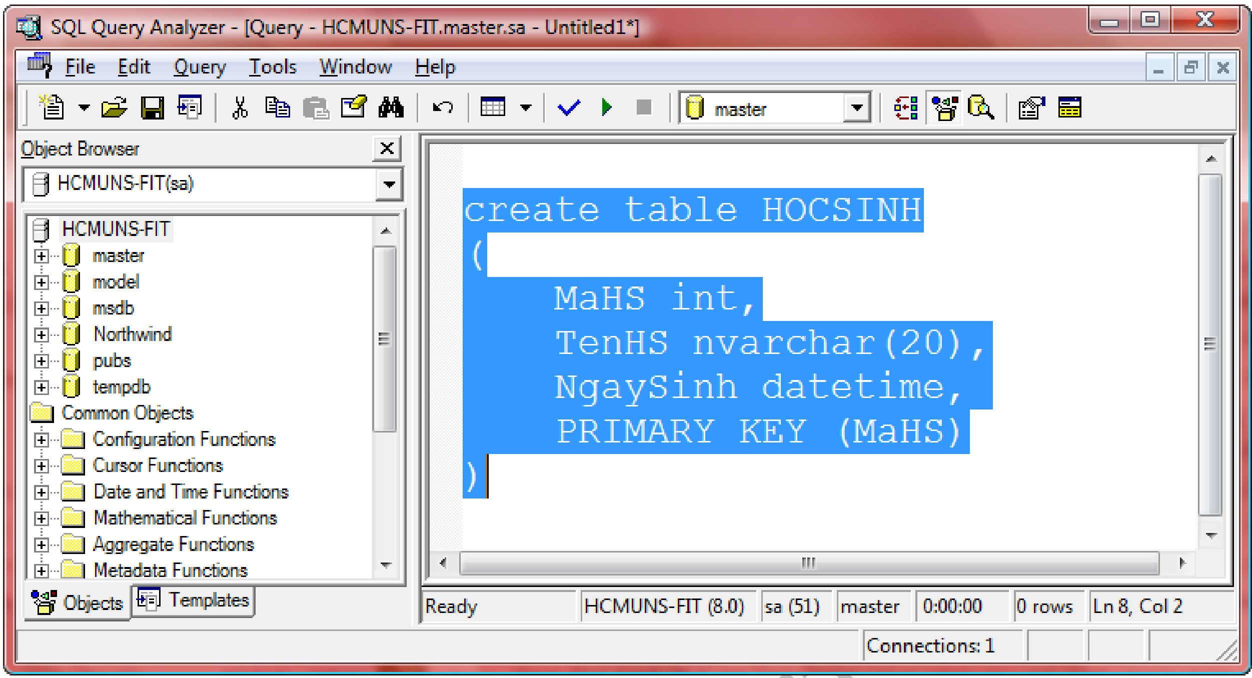

<u>Step 2</u>: Click the RUN button or press F5 to excute the selected.

<u>Bước 3</u>: View the returned output.

2.1.2 Creating a database

Syntax	**Example**
<u>Creating a database:</u> CREATE DATABASE [Database name] <u>Select the database to work with:</u> USE [Database name] <u>Delete database</u>: DROP DATABASE [Database name]	<u>Create the **HocSinhDB** database:</u> CREATE DATABASE HocSinhDB <u>Select the **HocSinhDB** database to work with</u> USE HocSinhDB <u>Delete the **HocSinhDB** database:</u> DROP DATABASE HocSinhDB

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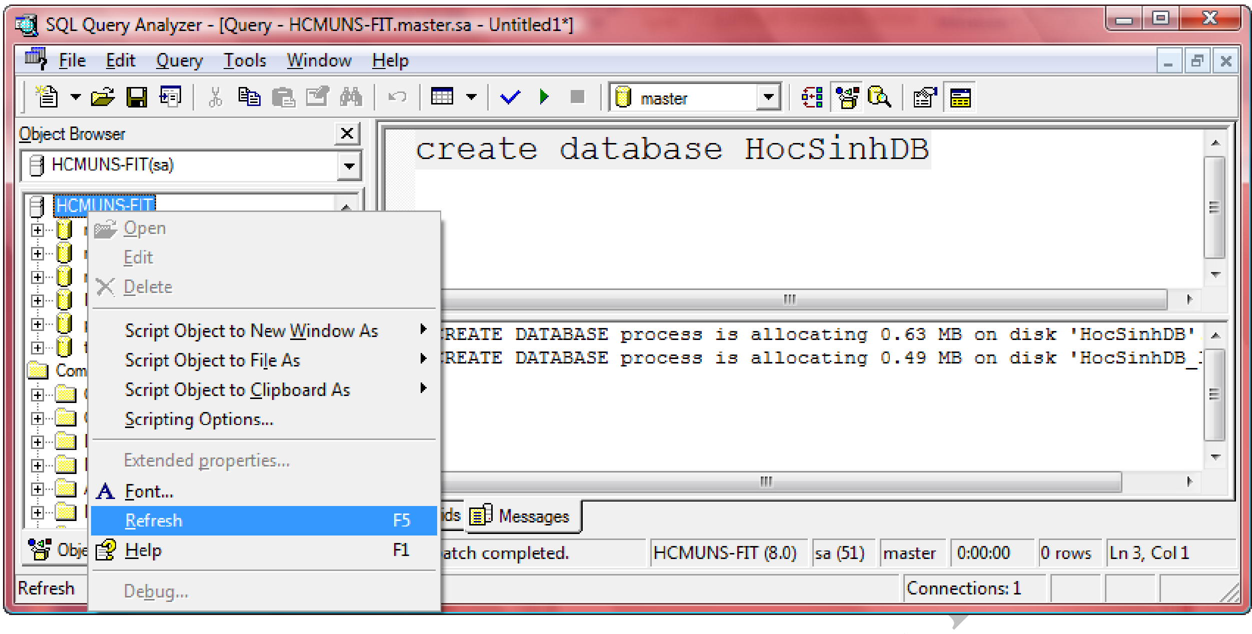

Notes:

1. After executing any statements that modify the database (e.g., create or drop), you must **REFRESH** the **Object Browser** window to see the latest changes.
2. To execute a **CREATE DATABASE** statement, the user must have the required permissions. For the accounts that students use in the computer lab, this permission is not granted if you log in using **Windows Authentication**, so you cannot run this command. Instead, log in to **SQL Server Management Studio** using the **username and password** provided by the instructors.

2.1.3 Create table

Common syntax for table creation:

Syntax	**Example**
CREATE TABLE [Table name] ([Attribute 1] [Data type 1], [Attribute 2] [Data type 2],	CREATE TABLE GIAOVIEN (MAGV char (5), HOTEN nvarchar(40),

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[Attribute 3] [Data type 3])	LUONG float, PHAI nchar(3), NGSINH datetime, MANQL char(5), MABM char(5))

Basic data types supported by SQL Server [1]:

#	Data	Data type
1	Integers	int, bigint, smallint, tinyint
2	Real (floating-point) numbers	float, real
3	Concurrency	money, smallmoney
4	Date time	datetime, smalldatetime
5	Strings	1 byte strings: char, varchar; 2 bytes strings (UNICODE): nchar, nvarchar
6	Binary strings	binary
7	Bit (1, 0)	bit
8	...	

[1] Students should consult Microsoft's documentation to learn about the valid domains of data types, as well as additional data types.

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The table-creation syntax above does not declare any integrity constraints for the table; therefore, it is typically used together with other statements (e.g., defining a primary key during table creation or defining foreign keys during table creation).

2.1.4 Drop table:

Syntax	**Example**
DROP TABLE [Table name]	<u>Drop the **GIAOVIEN** table</u>: DROP TABLE GIAOVIEN

<u>Notes:</u>

1. If the table already exists, or if the database contains an object with the same name as the table you want to create, the **CREATE TABLE** statement will fail.
2. If the table does not exist, the **DROP TABLE** statement will fail.
3. Nếu bảng đã tồn tại hoặc trong cơ sở dữ liệu có một đối tượng nào trùng tên với tên bảng muốn tạo thì câu lệnh tạo bảng sẽ bị lỗi.
4. Nếu bảng không tồn tại thì câu lệnh xóa bảng sẽ bị lỗi.

The two rules above also apply to all other database objects (e.g., primary keys, foreign keys, databases, etc.).

2.1.5 Defining (Creating) Primary Keys, Foreign Keys, and other Integrity Constraints

<u>Defining Primary Keys</u>:

Syntax	**Example**
<u>Define the primary key **inline** within **CREATE TABLE** statement (single-column PK):</u> CREATE TABLE [Table name]	<u>Create the **GIAOVIEN** table with inline primary key:</u> CREATE TABLE GIAOVIEN (

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([Attribute 1] [Data type 1] PRIMARY KEY, [Attribute 2] [Data type 2], [Attribute 3] [Data type 3],)	MAGV char (5) PRIMARY KEY, HOTEN nvarchar(40), LUONG float, PHAI nchar(3), NGSINH datetime, MANQL char(5), MABM char(5))

Syntax	**Example**
Define the primary key within the **CREATE TABLE** statement: CREATE TABLE [Table name] ([Attribute 1] [Data type 1], [Attribute 2] [Data type 2], [Attribute 3] [Data type 3], PRIMARY KEY ([The table's primary key attributes]))	Create the **GIAOVIEN** table with primary: CREATE TABLE GIAOVIEN (MAGV char (5), HOTEN nvarchar(40), LUONG float, PHAI nchar(3), NGSINH datetime, MANQL char(5), MABM char(5)

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	PRIMARY KEY (MAGV))

Syntax	**Example**
Define primary key after table creation: ALTER TABLE [Table name] ADD CONSTRAINT [Contraint name] PRIMARY KEY ([The table's primary key attributes]) **Notes**: When defining a primary key after the table creation, the primary key attributes (columns) must be declared as NOT NULL in the CREATE TABLE statement.	Define the primary key for the GIAOVIEN table: ALTER TABLE **GIAOVIEN** ADD CONSTRAINT **PK_GIAOVIEN** PRIMARY KEY (**MAGV**)

Defining Foreign Key:

Syntax	Example
Define foreign key after table creation: ALTER TABLE [Table name] ADD CONSTRAINT [Constraint name] FOREIGN KEY ([The table's primary key attributes]) REFERENCES [Referenced table name.] ([List of primary key columns	Thêm khóa ngoại cho bảng BOMON tham chiếu đến bảng KHOA: ALTER TABLE **BOMON** ADD CONSTRAINT FK_BOMON_KHOA FOREIGN KEY (MAKHOA) REFERENCES KHOA(MAKHOA)

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of the referenced table.])	

Define domain constraint:

Syntax	Example
ALTER TABLE [Table name] ADD CONSTRAINT [Constraint name] CHECK ([Conditional Expression])	Add a constraint requiring Phái to be either Nam or Nữ ALTER TABLE **GIAOVIEN** ADD CONSTRAINT **C_PHAI** CHECK (PHAI IN ('Nam', N'Nữ'))

Create a UNIQUE constraint (candidate key)[2]:

Syntax	Example
ALTER TABLE [Table name] ADD CONSTRAINT [Constraint name] UNIQUE ([List of attributes])	Add a constraint requiring the full name (Họ tên) to be unique. ALTER TABLE **GIAOVIEN** ADD CONSTRAINT **U_HOTEN** UNIQUE (HOTEN)

Drop a primary key, foreign key, or domain constraint:

Syntax	Example
ALTER TABLE [Table name] DROP CONSTRAINT [Constraint name]	Drop a primary key: ALTER TABLE **GIAOVIEN** DROP

[2] A unique constraint is also considered a key, and foreign keys can reference columns that have been defined with a UNIQUE constraint.

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	CONSTRAINT PK_GIAOVIEN Drop a foreign key: ALTER TABLE **BOMON** DROP CONSTRAINT FK_BOMON_KHOA

2.1.6 Add, modify, or drop table attributes

Add attributes:

Syntax	**Example**
ALTER TABLE [Table name] ADD [Attribute] [Data type]	Add the DIACHI attribute: ALTER TABLE **GIAOVIEN** ADD DIACHI nvarchar(20)

Drop attribute

Syntax	**Example**
ALTER TABLE [Table name] DROP COLUMN [Attribute]	Drop the DIACHI attribute ALTER TABLE **GIAOVIEN** DROP COLUMN DIACHI

Modify attribute:

Cú pháp	**Ví dụ**
ALTER TABLE [Table name] ALTER COLUMN [Attribute] [New data type]	Modify the DIACHI attribute ALTER TABLE GIAOVIEN ALTER COLUMN DiaChi nvarchar(100)

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2.1.7 Drop table

Cú pháp	Ví dụ
DROP TABLE [Table name]	Drop the GIAOVIEN table DROP TABLE **GIAOVIEN**

Dropping tables with foreign key relationships:

General rule: A table cannot be dropped if it is referenced by a foreign key constraint.

Therefore:

1. If there is **no circular dependency**, drop the **foreign-key (child) tables** first, then drop the referenced (parent) tables. Alternatively, drop the foreign key constraints first and then drop the tables.
2. If there is a **circular dependency**, drop one foreign key constraint to break the cycle, then follow the procedure in (1).

2.1.8 Useful Statements for Inspecting Metadata

Syntax	Example
Retrieve metadata information about a database object: sp_help [Table name]	sp_help GIAOVIEN
View primary key information: sp_pkeys [Table name]	sp_pkeys GIAOVIEN
View foreign key information: sp_fkeys [Table name]	sp_fkeys GIAOVIEN

2.1.9 Notes

– A **primary key may consist of multiple attributes**, which is called a **composite primary key**.

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- A table can define **at most one primary key**, but it may define **multiple candidate keys**.
- A **candidate key** is implemented as a **UNIQUE** constraint.
- A **foreign key** must reference a key. In the examples above, foreign keys reference primary keys; however, a foreign key may also reference a **candidate key (UNIQUE)**.
- Primary key and foreign key names are mainly **for readability**, but students should follow a consistent naming convention for clarity (e.g., primary key names start with **PK_**, foreign key names start with **FK_**).
- For composite primary keys or foreign keys, list the columns **separated by commas**.

2.2 Inserting and Updating Data

2.2.1 Syntax for Inserting a Single Row into a Table

Syntax	**Example**
<u>Implicit insert syntax</u>: INSERT INTO [Table name] VALUES ([val_1], [val_2], ..., [val_n]) Notes : The [val_1], [val_2], ..., [val_n]) correspond to the table's columns. Therefore, the user must know the column order in order to supply values that match the table structure defined in the **CREATE TABLE** statement.	Assume the table: **GIAOVIEN** (**<u>MAGV</u>**, HOTEN, NGSINH, LUONG) *<u>Insert one row into GIAOVIEN with **all values** provided</u>*: INSERT INTO GIAOVIEN VALUES ('**GV01**', 'Nguyen Van An', '12/1/2008', **10000**) *<u>Insert one row into GIAOVIEN where **LUONG is NULL**:</u>* INSERT INTO GIAOVIEN VALUES ('**GV02**', 'Tran Thi Be, '12/1/2008', **NULL**)
<u>Explicit insert syntax</u>: INSERT INTO [Table name] ([att_1], [att_2], ..., [att_n]) VALUES ([val_1], [val_2], ..., [val_n]) **<u>Lưu ý</u>**: The inserted values must correspond	*<u>Insert one row into GIAOVIEN with **all values** provided</u>*: INSERT INTO NHANVIEN (MAGV, HOTEN, NGSINH, LUONG) VALUES ('**NV03**', 'Nguyen Manh Hung', '12/1/2008', 40000)

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to the declared attributes.	***Insert one row into GIAOVIEN where LUONG is NULL:*** INSERT INTO GIAOVIEN (MAGV, HOTEN, NGSINH) VALUES ('**NV04**', 'Nguyen Manh Hung', '12/1/2008')
Insert data from an existing source: INSERT INTO ... SELECT ... **Key point:** This approach can insert multiple rows in a single statement.	

2.2.2 Inserting String and Date/Time Data

Syntax	Example
Insert Unicode string: Prefix Unicode string literals with N.	INSERT INTO GIAOVIEN VALUES ('NV01', **N'Nguyễn Văn Trường'**, '12/30/1955', 5000)
Insert date and time: Default Datetime input format : **'mm/dd/yyyy'**	INSERT INTO GIAOVIEN VALUES ('NV01', N'Nguyễn Văn Trường', **'12/30/1955'**, 5000)
Insert a row in which one value is NULL[3] Use NULL keyword	INSERT INTO GIAOVIEN VALUES ('NV01', 'Tran Thi Be', '12/1/2008', **NULL**)

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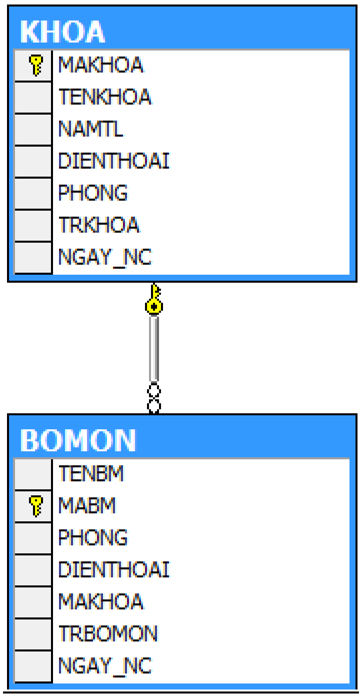
[3] If a column is declared as NOT NULL in the table definition, it must have a value when inserting a row into the table.

2.2.3 Inserting Data with Foreign Key Constraints

Case 1:



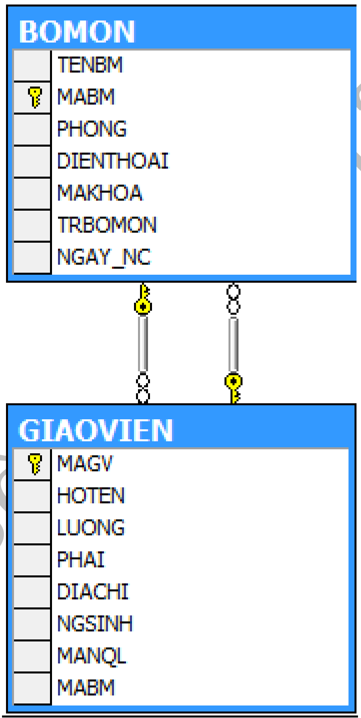
Approach 1:

1. Step 1 : Input KHOA
2. Step 2 : Input BOMON

Approach 2:

1. Step 1 : Input BOMON with **MAKHOA = NULL**
2. Step 2 : Input KHOA
3. Step 3 : Update the **MAKHOA** attribute in the BOMON table

Case 2:



Approach 1:

Step 1 : Input GIAOVIEN, with **MABM** = **NULL**

Step 2 : Input BOMON

Step 3 : Update the **MABM** attribute inside GIAOVIEN

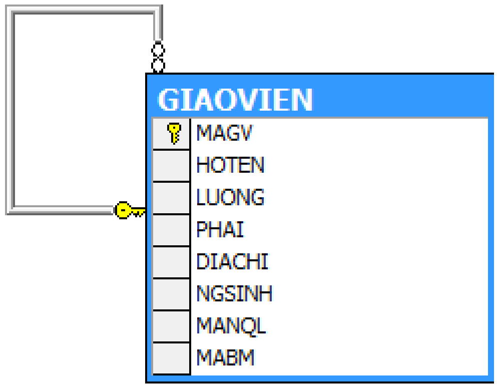
Approach 2 :

Step 1 : Input BOMON, with **TRBOMON** = **NULL**

Step 2 : Input GIAOVIEN

Step 3 : Update the **TRBOMON** attribute inside BOMON

Case 3:

	*Approach 1 :* Step 1: First, insert the **GIAOVIEN** records whose **MANQL** is NULL. Step 2: Then, insert the GIAOVIEN records for which the manager (NQL) information has already been inserted. *Approach 2 :* Step 1. Insert all GIAOVIEN records, setting MANQL to NULL Step 2. Update MANQL for GIAOVIEN.

2.2.4 Viewing and Deleting Data

Syntax	**Example**
Retrieve all data from a table: SELECT * FROM [Tên bảng]	*View all data of the GIAOVIEN table:* SELECT * FROM GIAOVIEN
Delete data froma table: DELETE FROM [Table name] WHERE [Condition]	*Delete the teacher with id NV01 from the **GIAOVIEN** table:* DELETE FROM **GIAOVIEN** WHERE MAGV = 'GV01' *Delete all data from the **GIAOVIEN** table:* DELETE FROM NHANVIEN

2.2.5 Updating data

Syntax	**Example**
UPDATE TABLE [Table name] SET $[att_1] = [val_1], [att_2] = [val_2], ..., [att_n] = [val_n]$ WHERE ([Condition]) **Notes:** If the WHERE clause is omitted, **all**	*Increase the salary of all teachers with a salary below 50,000 by 10%:* UPDATE TABLE **GIAOVIEN** SET LUONG=LUONG * 1.1

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rows in the table will be updated.	WHERE LUONG < 50000 Update the name and date of birth of the teacher with **MAGV = '001'** to **'Hùng'** and **'1/1/1984'**: UPDATE TABLE GIAOVIEN SET HOTEN = **N**'Hùng', NGSINH='1/1/1984' WHERE MAGV='001'

2.2.6 Best Practices

In practice, the workflow for building a database schema and loading data is usually as follows:

Step 1: Create tables and define **primary key** constraints.

Step 2: Define **foreign key** constraints and other constraints.

Step 3: Insert data.

2.2.7 Creating a Database Diagram Using Enterprise Manager

Purpose: To view tables and their relationships in a visual (diagram) form.

Step 1: Right-click and select **New Database Diagram**, as shown in the figure.

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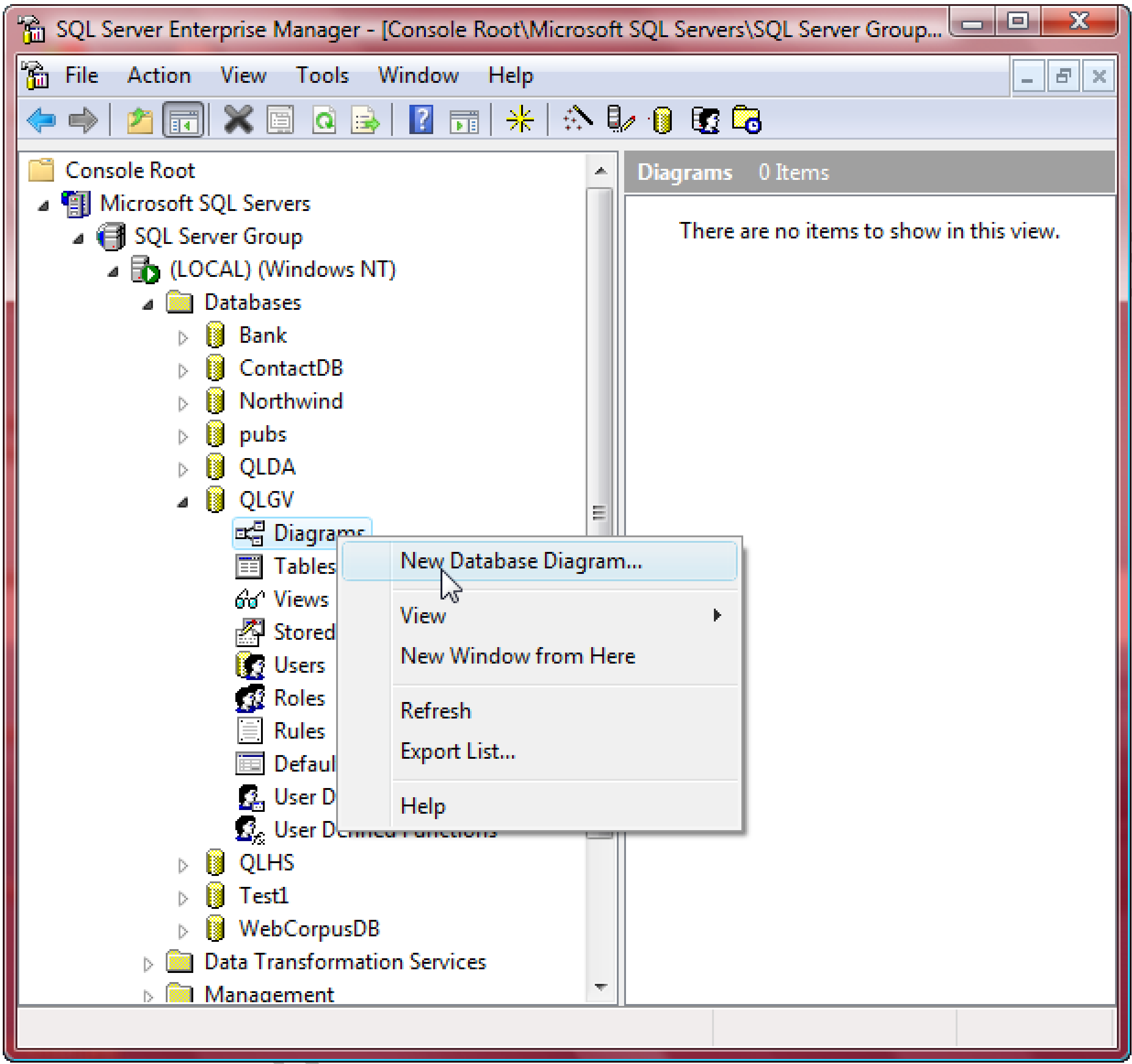

<u>Step 2</u>: Select **Next.**

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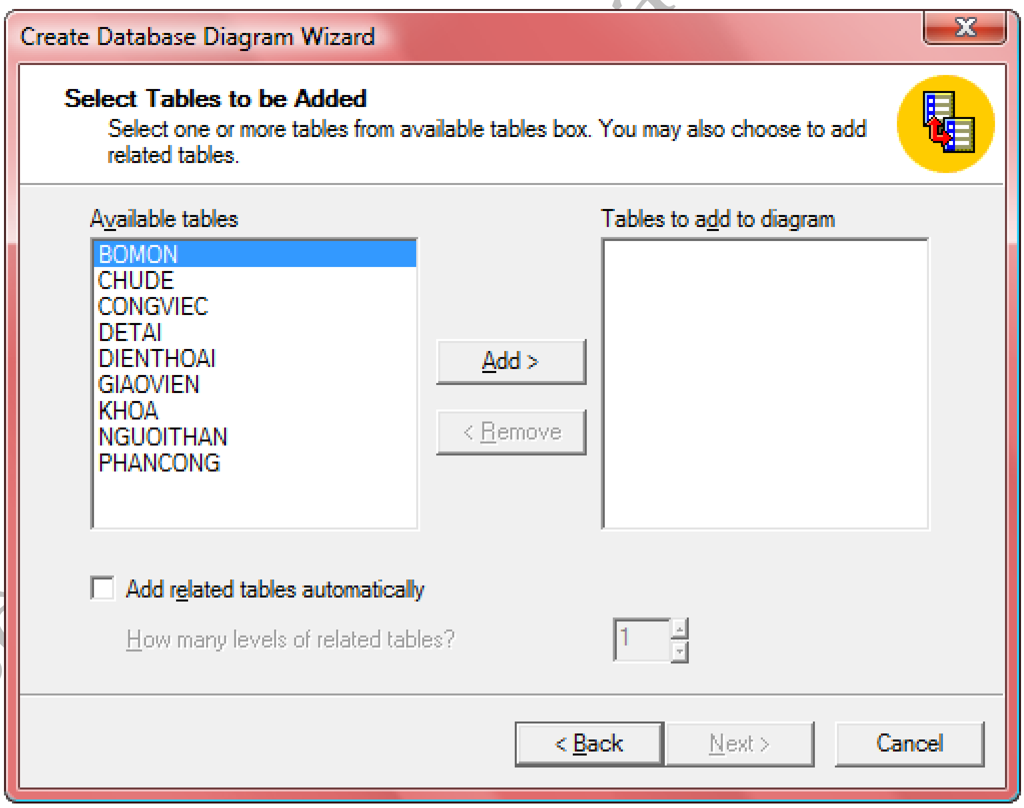
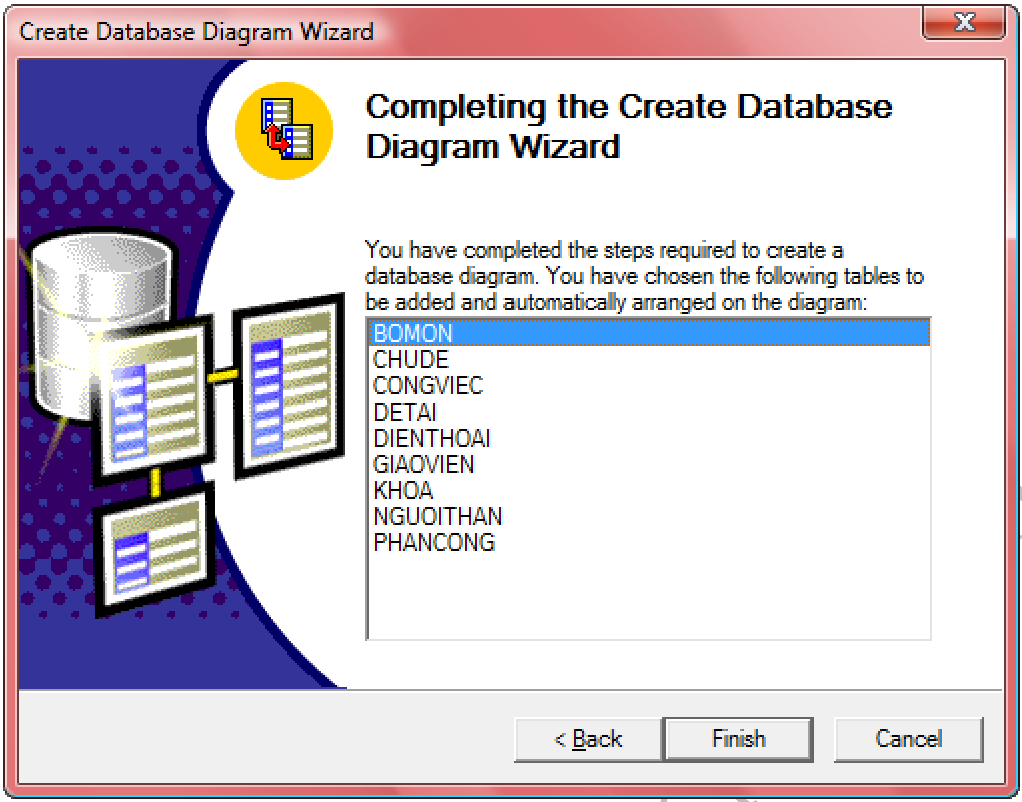
Step 3: Select (**Add**) the tables to display in the diagram.



Step 4: Select Finish.



Step 5: View the diagram.

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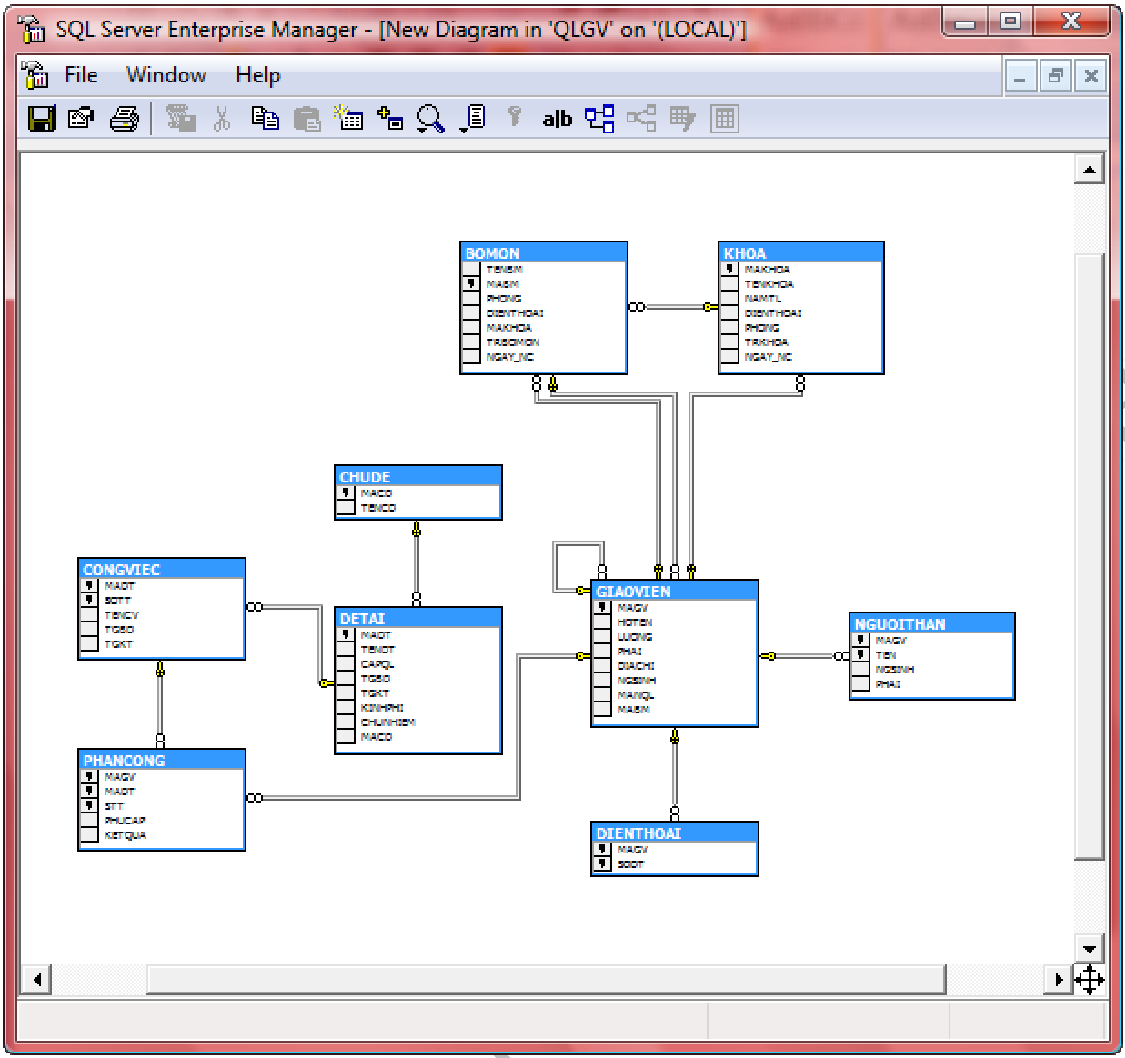

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3 In-Class Exercises

Requirement**:** Write a script to create the schema and insert data for the **GIAOVIEN**, **BOMON**, and **KHOA** tables in the Quan ly DE TAI exercise.

Duration: 1 hour.

4 Homework Exercises

Requirement**:**

Complete the script to create the schema and insert data for the Quan ly DE TAI database.

Duration: 3 hours.

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